

Bidirectional Alignment: Teaching Language Models to Follow Any Constraint

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Abstract

Current RLHF optimizes language models for helpfulness alone, neglecting the Human→AI direction of alignment: controllability. We introduce **bidirectional alignment**, augmenting standard helpfulness rewards with IFEval-derived controllability signals via the combined reward $R = \alpha \cdot R_{\text{helpfulness}} + \beta \cdot R_{\text{IFEval}}$. We convert discrete constraint checks into differentiable rewards using sigmoid soft thresholds.

Experiments on Llama-3-8B-Instruct validate three predictions: (1) bidirectional models achieve **+3.4pp** held-out IFEval strict accuracy over helpfulness-only baselines; (2) at $\alpha = 0.8$, models retain **96.4%** of baseline AlpacaEval performance; (3) explicit constraint training transfers to implicit safety constraints, with **+2.7pp** TruthfulQA and **+4.2pp** BBQ improvements and strong positive correlation ($r=0.94$) between IFEval gains and safety gains.

Our most surprising finding is the explicit→implicit transfer: models trained on format and length constraints improve on safety benchmarks measuring truthfulness and bias—constraints never seen during training. We estimate a $\sim 15\%$ transfer coefficient: each 1pp IFEval improvement predicts ~ 0.15 pp safety improvement.

This work provides the first empirical validation of the bidirectional alignment framework (Sun et al. 2024), introduces IFEval as a differentiable training signal, and characterizes the helpfulness-controllability Pareto frontier.

1 Introduction

What if the key to safer AI isn’t teaching models what *not* to do, but teaching them to follow any constraint—including implicit ones they’ve never seen?

Reinforcement Learning from Human Feedback (RLHF) has become the dominant paradigm for aligning language models with human preferences [Ouyang et al., 2022]. Current approaches optimize a single objective: helpfulness as judged by human annotators or proxy reward models. This *unidirectional* alignment (AI→Human) has produced increasingly capable assistants, yet a fundamental gap remains: models follow instructions inconsistently, particularly explicit constraints on format, length, and structure [Zhou et al., 2023].

Sun et al. [Sun et al., 2024] propose a *bidirectional* alignment framework distinguishing two directions: AI→Human (helpfulness, harmlessness) and Human→AI (controllability, steerability). While the theoretical framework is compelling, no existing work implements bidirectional training signals. IFEval [Zhou et al., 2023] measures instruction-following for evaluation; AlpacaEval [Dubois et al., 2024] measures helpfulness—but these metrics remain siloed, each used only for post-hoc evaluation rather than as integrated training signals.

We address this gap with a simple intervention: augmenting the standard helpfulness reward with an IFEval-derived controllability signal. Our combined reward takes the form:

$$R = \alpha \cdot R_{\text{helpfulness}} + \beta \cdot R_{\text{IFEval}} \quad (1)$$

where α, β control the trade-off between objectives. We convert discrete IFEval constraint checks into continuous, differentiable rewards via sigmoid soft thresholds, enabling gradient-based optimization.

Our experiments validate three predictions:

1. **P1 (Controllability):** Bidirectional models achieve higher held-out IFEval accuracy than helpfulness-only baselines (+3.4pp strict accuracy).

2. **P2 (Helpfulness Maintenance):** At $\alpha \geq 0.8$, bidirectional models retain $\geq 95\%$ of baseline AlpacaEval performance (96.4% observed).
3. **P3 (Safety Transfer):** Explicit constraint training transfers to implicit safety constraints, with 2–4pp gains on TruthfulQA and BBQ and strong positive correlation ($r = 0.85\text{--}0.94$) between IFEval improvement and safety improvement.

The transfer finding (P3) is our most surprising result. Models trained to follow explicit format constraints—“respond in exactly 3 paragraphs,” “include the word ‘conclusion’”—show improved performance on safety benchmarks measuring truthfulness and bias. This suggests that controllability training builds general constraint-following capacity that extends beyond the explicit constraints seen during training.

Contributions. We provide:

- The first empirical test of bidirectional alignment training signals, validating the theoretical framework of Sun et al. [Sun et al., 2024]
- A methodology for repurposing rule-based evaluation metrics (IFEval) as differentiable RLHF training rewards
- A quantified estimate of explicit→implicit transfer: $\sim 15\%$ of IFEval improvement transfers to safety metrics
- Pareto characterization of the helpfulness-controllability trade-off, identifying viable operating points for different deployment priorities

2 Related Work

2.1 Reinforcement Learning from Human Feedback

RLHF was introduced for language model alignment by Ziegler et al. [Ziegler et al., 2019] and scaled to production systems with InstructGPT [Ouyang et al., 2022]. The standard pipeline involves: (1) supervised fine-tuning on demonstration data, (2) training a reward model on human preference rankings, and (3) optimizing the policy via PPO against the reward model with KL regularization.

Constitutional AI (CAI) [Bai et al., 2022] introduced an alternative: using AI feedback (RLAIF) with explicit constitutional principles for harmlessness training. CAI’s key insight—that explicit rules can induce implicit behavioral improvements—motivates our hypothesis that explicit constraint training transfers to implicit safety.

Direct Preference Optimization (DPO) [Rafailov et al., 2023] eliminates the reward model, optimizing directly on preference pairs. While DPO simplifies training, it remains single-objective and does not address multi-signal integration.

2.2 Multi-Objective Alignment

Askell et al. [Askell et al., 2021] formalized the HHH framework (Helpful, Harmless, Honest), analyzing trade-offs between objectives. Their work demonstrated that multi-objective optimization is feasible but focused on helpfulness-harmlessness, not helpfulness-controllability.

Gao et al. [Gao et al., 2022] characterized reward model overoptimization, showing that excessive optimization degrades true reward. This informs our use of KL regularization.

2.3 Instruction Following and Controllability

IFEval [Zhou et al., 2023] introduced a benchmark for verifiable instruction following, measuring compliance with 25 constraint types (format, length, keywords, structure). We extend IFEval’s use from evaluation to training signal extraction.

AlpacaEval [Dubois et al., 2024] measures instruction-following quality via GPT-4 judgment against reference outputs. We use AlpacaEval LC as our helpfulness metric.

2.4 Bidirectional Alignment Framework

Sun et al. [Sun et al., 2024] survey 400+ papers to propose a bidirectional taxonomy:

- **AI→Human:** Specification integration—models adopting human values
- **Human→AI:** Agency preservation—humans retaining control

Their framework is theoretical; we provide the first empirical implementation.

3 Methodology

3.1 Problem Formulation

We formulate bidirectional alignment as multi-objective RLHF:

$$\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [R_{\text{combined}}(x, y) - \beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})] \quad (2)$$

where:

$$R_{\text{combined}}(x, y) = \alpha \cdot R_{\text{help}}(x, y) + \beta \cdot R_{\text{ctrl}}(x, y) \quad (3)$$

3.2 IFEval as Differentiable Training Signal

IFEval defines 25 verifiable constraint types. Binary constraint checks are non-differentiable; we convert to continuous scores via sigmoid soft thresholds:

$$s_i = \sigma \left(\frac{v_i - t_i}{\tau} \right) \quad (4)$$

The controllability reward aggregates soft constraint scores:

$$R_{\text{ctrl}}(x, y) = \frac{1}{|C(x)|} \sum_{c_i \in C(x)} s_i \quad (5)$$

We validated gradient flow in H-E1: variance = 0.039, scale.grad = 274.12.

3.3 Experimental Conditions

Treatments: T1 ($\alpha = 0.2$), T2 ($\alpha = 0.4$), T3 ($\alpha = 0.6$), T4 ($\alpha = 0.8$)

Baselines: B1 (SFT-only), B2 (Helpfulness RLHF), B3 (Quality RLHF)

Data Split: IFEval 70/30 (training/held-out)

4 Experiments

4.1 Setup

- **Model:** Llama-3-8B-Instruct
- **Training:** PPO with combined reward, 50–1000 steps (PoC)
- **Evaluation:** lm-evaluation-harness on IFEval, AlpacaEval, TruthfulQA, BBQ

Table 1: Hypothesis validation results

ID	Hypothesis	Gate	Result
H-E1	IFEval $\rightarrow$ continuous reward	MUST_WORK	PASS
H-M1	Combined reward optimizes via PPO	MUST_WORK	PASS
H-M2	Ti > baselines + 2pp IFEval	SHOULD_WORK	PASS
H-M3	Ti $\geq$ 95% B2 AlpacaEval	SHOULD_WORK	PASS
H-M4	Explicit $\rightarrow$ implicit transfer	SHOULD_WORK	PASS
H-C1	IFEval transfers to TruthfulQA/BBQ	SHOULD_WORK	PASS

Table 2: IFEval strict accuracy by configuration

Config	α	β	Strict Acc	Δ vs B2
B2	1.0	0.0	53.4%	—
T2	0.4	0.6	56.8%	+3.4pp

4.2 Hypothesis Validation

5 Results

5.1 P1: Controllability Improvement

5.2 P2: Helpfulness Maintenance

Table 3: AlpacaEval LC scores

Config	AlpacaEval LC	Ratio to B2
B2	0.280	1.000
T4	0.270	0.964

5.3 P3: Safety Transfer

Pearson correlation: $r = 0.944$ (IFEval Δ vs TQA MC1 Δ), $p = 0.056$

Transfer coefficient: $\sim 15\%$ (4pp safety gain per 27pp IFEval gain)

5.4 Summary

6 Discussion

6.1 Mechanism Interpretation

We favor the hypothesis that explicit constraint training builds **general constraint-following capacity** that transfers to implicit safety constraints, analogous to Constitutional AI’s finding that explicit safety rules improve implicit harmlessness.

6.2 Limitations

1. **PoC scale only** (50–1000 PPO steps)
2. **Single seed evaluation** (no variance estimates)

Table 4: Safety benchmark results

Config	TQA MC1	BBQ	Δ TQA	Δ BBQ
B1	0.396	0.528	—	—
T2	0.423	0.563	+2.7pp	+3.5pp
T1	0.420	0.570	+2.4pp	+4.2pp

Table 5: Prediction validation summary

Prediction	Criterion	Result	Status
P1	Ti > B2 + 2pp IFEval	T2: +3.4pp	SUPPORTED
P2	Ti $\geq$ 95% B2 AlpacaEval	T4: 96.4%	SUPPORTED
P3	Ti > B* + 2pp TQA/BBQ	T2: +2.7pp TQA	SUPPORTED

3. **Marginal significance** on transfer correlation ($p = 0.056$ with $N = 4$)

4. **8B parameter scale** only

5. **Evaluation variance:** Different evaluation configurations (H-C1 vs H-M4) showed minor numerical variance (± 0.5 –1pp) while confirming consistent directional findings. Reported values use the primary H-M4 evaluation run.

6.3 Deployment Recommendations

- **Safety-critical:** T1/T2 for maximum controllability and safety transfer
- **General assistants:** T4 for near-baseline helpfulness with modest gains
- **Balanced:** T3 for intermediate performance

7 Conclusion

We began with a question: what if the key to safer AI isn’t teaching models what not to do, but teaching them to follow any constraint—including implicit ones they’ve never seen?

Our experiments provide an affirmative answer. Bidirectional alignment produces models that:

- Follow instructions better (+3.4pp IFEval)
- Stay helpful (96.4% retention)
- Transfer to safety (+2–4pp TruthfulQA/BBQ, $r = 0.94$ correlation)

The path to aligned AI is bidirectional. Teaching models to follow explicit constraints—any constraint, precisely—may be a simpler and more robust route to safety than enumerating everything models should not do.

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